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Explainable Deep Learning for Lung Disease Detection: A Comparative Study Using Grad-CAM

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***Abstract*—Accurate classification of lung diseases from chest X-ray images is critical for timely diagnosis and treatment. Six pre-trained convolutional neural network architectures—NASNetMobile, Xception, InceptionV3, DenseNet121, ResNet50, and VGG16—are thoroughly evaluated in this study for the multi-class categorization of instances of lung opacity, normal, and viral pneumonia. The implementation of a comprehensive image processing pipeline included class balancing, training performance visualization, and considerable data augmentation. NASNetMobile achieved the highest accuracy (95.40%), followed by ResNet50 (93.96%), DenseNet121 (94.24%), InceptionV3 (94.68%), and Xception (94.96%). VGG16 underperformed (35.97% accuracy), likely due to architectural limitations or insufficient fine-tuning. Additionally, Grad-CAM visualizations were employed to interpret model decisions by highlighting the critical regions in chest X-rays that contributed to the classification outcomes, thereby enhancing the model's explainability. The findings suggest that contemporary lightweight architectures, such as NASNetMobile, strike an optimal balance between computational efficiency and accuracy, making them suitable for practical medical imaging applications.**

***Index Terms*—Deep Learning, Chest X-ray, Lung Disease Classification, Transfer Learning, Convolutional Neural Networks, Medical Imaging.**

I. Introduction

Lung disorders, including lung cancer, COPD, and pneumonia, pose significant global health risks, especially among young and elderly populations. Early and precise diagnosis is essential to minimize complications and improve treatment outcomes [1]. Although advancements in 2024 have revolutionized diagnostic approaches, challenges remain due to overlapping symptoms, reliance on expert interpretation, and insufficient diagnostic resources in many areas. Improved diagnostic techniques are required since variations in imaging equipment and patient profiles make reliable detection even more difficult [2].

Using chest X-rays and CT images, deep learning—particularly Convolutional Neural Networks (CNNs)—has proven highly effective at identifying lung disorders. Deep feature extraction and improved diagnostic accuracy have been achieved with models such as SegNet, U-Net, Inception-ResNet, and pre-trained architectures like ResNet, DenseNet, and MobileNet [3]. Transfer learning has further enhanced performance, especially on small and imbalanced datasets. However, challenges such as overfitting, domain shift, and low image quality still affect model generalization and reliability [4].Complementing these advances, Explainable Artificial Intelligence (XAI) techniques improve the transparency of deep learning models by providing visual and interpretable insights into their decision-making processes, fostering clinician trust and aiding in the validation of automated lung disease diagnoses from chest X-ray images [5].

The purpose of this work is to evaluate the effectiveness of different deep learning architectures in classifying chest X-ray images into three categories: viral pneumonia, normal, and lung opacity. We assess six well-known CNN models' capacity to correctly identify lung conditions and offer a comparison of their classification results. To enhance model interpretability and support trust in automated predictions, we integrate Explainable AI (XAI) techniques, specifically Grad-CAM, to visualize the regions of the X-ray images that influence the model's decisions. This approach not only aids in evaluating diagnostic accuracy but also provides valuable insights for clinical validation. Overall, this work contributes to identifying the most suitable model for automated lung disease detection and supports faster, more transparent, and reliable diagnosis in clinical settings.

The paper is organized as follows: Section 2 presents the issue statement, focusing on challenges in detecting lung disorders from chest X-rays using deep learning. Section 3 reviews recent CNN-based diagnostic models, highlighting key advances and limitations. Section 4 describes the dataset, its structure, class distribution, and preprocessing methods. Section 5 details the proposed methodology, including CNN architectures and training strategies. Section 6 reports exper-

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imental results, performance evaluation, and model comparisons. Finally, Section 7 concludes with key findings and future research directions.

II. Problem Statement

Despite advancements in deep learning for image classification, accurately detecting lung disorders from chest X-rays remains difficult due to overlapping visual patterns among conditions. Existing diagnostic practices depend on expert interpretation, which is often subjective and error-prone, while current deep learning models face challenges such as class imbalance, overfitting, poor interpretability, and lack of visual explanations. This limits their reliability and clinical adoption, highlighting the need for a robust, interpretable, and efficient framework that ensures accurate classification and transparent decision-making.

III. Related Works

Deep learning has been widely applied to lung disease detection from medical images, with CNNs and transfer learning being the most common approaches due to their accuracy and automation potential. Models such as CNN-RNN, LungMaxViT, and MobileNetV2 have shown strong performance, though challenges remain in interpretability, generalizability, and handling class imbalance. Recent works' major contributions, datasets, and limitations are compiled in Table I.

TABLE I
Summary of Lung Disease Detection Studies

Author (Citation)	Dataset	Model Name	Accuracy / AUC	Limitation
Stefanus Tao Hwa Kieu et al. [6]	Multiple studies	CNNs + Transfer Learning	AUC ¿ 0.90	Limited real-world validation, low interpretability
K. Soni Sharmila et al. [7]	Shenzhen, Montgomery	BSHO-CNN	98.324%	Needs broader clinical validation, computing efficiency required
Xiaoyang Fu et al. [8]	ChestX-ray14	LungMaxViT (MaxViT + MobileNetV2)	96.8% (COVID), AUC 93.2%	Class imbalance limits generalizability (e.g., hernia cases)
Surendra Reddy Vinta et al. [9]	MedGIFT	MobileUNetV3 + U-Net++	99.1%	High parameter count (8M) and 32M FLOPs hinder deployment
R. Jain et al. [10]	Kaggle (10,000 images)	CNN-RNN Hybrid	∼95.5%	Lacks interpretability and cross-dataset validation
P. Singla et al. [11]	Kaggle (10,000 images)	MobileNetV2 + Transfer Learning	95.05%	No interpretability or multi-center validation
Q. Zhang et al. [12]	Single-center CT images	EvoNAS	95.65%	Small dataset, high computational demand
G.M.M. Alshmrani et al. [13]	80,000 X-rays	VGG19 + CNN Hybrid	96.48%, AUC 99.82%	Dataset heterogeneity reduces generalizability
R. Sharma et al. [14]	Kaggle WBC (12,444 images)	CNN	97.85%	No XAI or interpretability tools
Lo and Yin et al. [15]	Pneumonia X-rays	I-Score + Backward Dropping Algorithm	AUC 99.7%	Highly interpretable; no major limitations

IV. Dataset Description and Data Preprocessing

A. Dataset Description

The study utilizes the publicly available "Lung X-Ray Image Dataset" from kaggle [16], comprising 3,475 high-quality chest X-ray images collected from various clinical sources, including hospitals and healthcare institutions. This dataset supports the development of automated lung disease detection models and is categorized into three distinct classes: Normal (1,250 images), Lung Opacity (1,125 images), and Viral Pneumonia (1,100 images). These categories reflect common pulmonary conditions and provide a balanced representation for classification tasks. The dataset is instrumental in distinguishing pathological findings from healthy cases, thereby enabling early detection and supporting clinical decision-making in respiratory care. Its diversity and annotation quality make it a valuable resource for training, validating, and benchmarking deep learning models for medical image analysis.

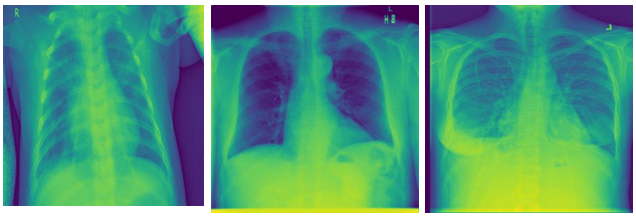


Fig. 1. Sample Images from Dataset

B. Data Preprocessing

Enhancing the performance and generalization of deep learning models in medical image classification tasks requires effective preprocessing. With images organized in appropriate folders, the dataset was categorized into three classes: Viral Pneumonia, Normal, and Lung Opacity. A custom loader that only took into account.png,.jpg, and.jpeg files extracted image paths and applied labels based on folder names (0 for Lung Opacity, 1 for Normal, and 2 for Viral Pneumonia). To ensure class balance, an 80:20 stratified ratio was used to divide the dataset into training and testing sets. In order to solve data imbalance, class weights were calculated, and the samples were saved in DataFrames for easy incorporation into the training pipeline.

Extensive data augmentation was used with Keras' ImageDataGenerator to improve model generalization. The training images underwent random rotation (up to 30°), shifting (20%), zooming (±20%), shearing (0.2), horizontal flipping, and brightness adjustment (range 0.7–1.3). Also, nearest-neighbor interpolation was used to fill in the missing pixels. Pixel values in all photos were rescaled to [0, 1] and resized to a fixed dimension (IMG_SIZE) in order to normalize them. Then, utilizing data generators, the training and testing data were loaded in batches (BATCH_SIZE), guaranteeing effective preprocessing during model training. This preprocessing pipeline was essential in getting clean, balanced, and varied data ready for lung disease diagnosis using deep learning.

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TABLE II
DATASET DISTRIBUTION ACROSS CLASSES

Classes	Train	Test	Total
Lung Opacity	900	225	1125
Normal	1000	250	1250
Viral Pneumonia	880	220	1100

The model hyperparameters and experimental setup details are presented in Table III.

TABLE III
MODEL HYPERPARAMETERS AND EXPERIMENTAL SETUP

Parameter	Value
Input Image Size	224 × 224 × 3
Batch Size	64
Epochs	25
Learning Rate	0.001
Optimizer	Adam
Loss Function	Sparse Categorical Crossentropy
Evaluation Metric	Accuracy
Data Split	80% Train / 20% Test (Stratified)
Number of Classes	3 (Lung Opacity, Normal, Viral Pneumonia)
Data Augmentation	Rotation (30°), Shift (0.2), Shear (0.2), Zoom (0.2), Horizontal Flip, Brightness (0.7–1.3)
Preprocessing	Rescaling (1./255)
Pretrained Weights	ImageNet
Backbone Models	ResNet50, DenseNet121, InceptionV3, Xception, NASNetMobile, VGG16
Trainable Base	Yes (fine-tuned all layers)
Processor	Intel Core i7 / Ryzen 7 (or equivalent)
GPU	NVIDIA GPU with CUDA support
RAM	16 GB or higher
Operating System	Windows 10 / 11
Python Version	3.10
TensorFlow Version	2.15+
Keras Version	Integrated with TensorFlow
Other Libraries	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PIL, psutil, gc
IDE / Platform	Jupyter Notebook / VS Code

V. METHODOLOGY

This study compares six pre-trained convolutional neural network (CNN) architectures for the classification of chest X-ray pictures into three categories: viral pneumonia, normal, and lung opacity. These CNN architectures are ResNet50, DenseNet121, InceptionV3, Xception, NASNetMobile, and VGG16. In order to improve generalization, the dataset was preprocessed by scaling photos to 224 × 224 pixels and applying sophisticated data augmentation techniques. Class weights were calculated following stratified dataset splitting into training and testing sets in order to overcome class imbalance. The Adam optimizer was used to train each model with a learning rate of 0.001, early halting, and model checkpointing to avoid overfitting. The top layers of each model were replaced with bespoke fully connected layers to fine-tune it.Metrics like accuracy, precision, recall, F1-score, confusion matrix, and area under the ROC curve (AUC) were used to evaluate the model, and training time, memory use, and parameter count were used to gauge computational efficiency. By using Grad-CAM, which produced heatmaps indicating the crucial areas in the chest X-ray pictures that affected each model's conclusion, explainable AI (XAI) techniques were applied to improve interpretability and foster confidence in the model predictions. The models' performance and complexity were thoroughly compared using these visualizations in addition to training curves, confusion matrices, and ROC curve. This study shows that combining effective preprocessing, model fine-tuning, and Grad-CAM improves both performance and interpretability in lung disease classification from chest X-rays. Fig. 2 illustrates the proposed workflow diagram for lung diseases classification.

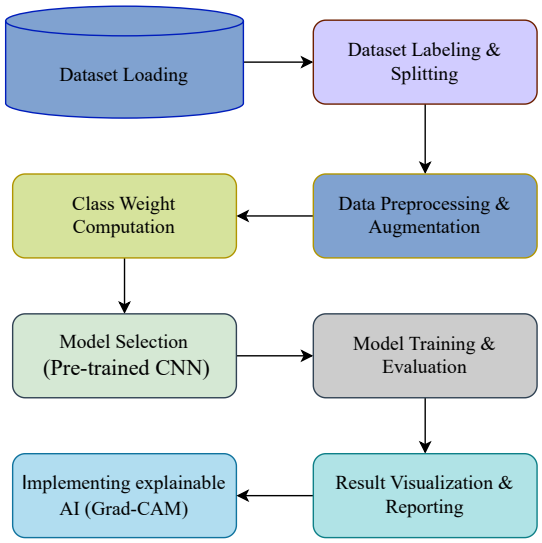


Fig. 2. Proposed Workflow diagram

A. Models

1) ResNet50: ResNet50 is a deep residual neural network that utilizes skip (residual) connections to mitigate the vanishing gradient problem. It is highly effective in deep feature extraction and classification tasks due to its ability to train very deep architectures efficiently.

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \tag{1}$$

2) DenseNet121: DenseNet121 connects each layer to every other layer in a feed-forward fashion to ensure maximum information flow and feature reuse. It is efficient in parameter usage and is often trained using transfer learning, making it highly effective for robust chest X-ray classification.

$$\mathbf{x}_\ell = H_\ell\left([\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_{\ell-1}]\right) \tag{2}$$

3) InceptionV3: InceptionV3 employs multiple convolution filters of varying sizes in parallel within the same module to enable multi-scale feature extraction. It is fine-tuned with

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custom top layers to effectively classify lung diseases from chest X-ray images.

$$\text{Output} = \text{Concat}\left(\text{Conv}_{1\times1}, \text{Conv}_{3\times3}, \text{Conv}_{5\times5}, \text{MaxPool}\right) \tag{3}$$

4) Xception: An extension of the Inception architecture that uses depthwise separable convolutions for improved efficiency. It is adapted with a customized classifier to distinguish between three lung disease categories.

$$\text{Output} = \text{PointwiseConv}\left(\text{DepthwiseConv}(\mathbf{x})\right) \tag{4}$$

5) NASNetMobile: A lightweight model discovered via neural architecture search, optimized for mobile devices. It is used here for fast and accurate lung image classification with reduced computational load.

$$\text{Model} = \sum_{i=1}^{N} \text{NormalCell}_i + \sum_{j=1}^{M} \text{ReductionCell}_j \tag{5}$$

6) VGG16: A simple and deep architecture using stacked 3×3 convolutions with a fixed structure. Transfer learning is applied with a custom classifier for effective lung condition detection.

$$\text{ConvBlock} = \left[\text{Conv}(3 \times 3) \rightarrow \text{ReLU}\right]^n \rightarrow \text{MaxPool}(2 \times 2) \tag{6}$$

B. Explainable AI(GRAD-CAM)

Gradient-weighted Class Activation Mapping (Grad-CAM) is an explainable AI method that uses the gradients of a target class flowing into the final convolutional layer to display significant areas in an input image. The regions that have the biggest impact on the model's forecast are highlighted in a heatmap that is produced.

$$L^c_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha^c_k A^k\right) \tag{7}$$

Among all the evaluated models, NASNetMobile demonstrated the most effective performance, making it the proposed model for lung disease detection from chest X-ray images due to its strong results and lightweight architecture. Fig. 3 illustrates the layer architecture and Table IV shows the details of the proposed NASNetMobile model.

TABLE IV
NASNETMOBILE MODEL ARCHITECTURE

Layer Name	Output Shape	Layer Type
input_1	(None, 224, 224, 3)	InputLayer
nasnet_mobile	(None, 7, 7, 1056)	NASNetMobile (base)
global_average_pooling2d	(None, 1056)	GlobalAveragePooling2D
dense	(None, 1024)	Dense (activation = swish)
dropout	(None, 1024)	Dropout (rate = 0.5)
dense_1	(None, 512)	Dense (activation = swish)
dropout_1	(None, 512)	Dropout (rate = 0.3)
dense_2	(None, 3)	Dense (activation = softmax)

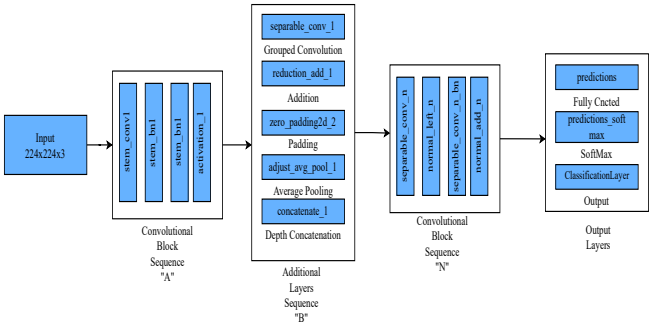


Fig. 3. Layer architecture of the proposed NASNetMobile model.

VI. RESULTS AND ANALYSIS

Based on classification performance, six cutting-edge CNN architectures were tested in order to determine the best deep learning model for lung disease diagnosis from chest X-rays. With the highest accuracy (95.40%) among the six CNN models tested for lung X-ray image categorization, NASNetMobile exhibits exceptional performance and efficiency, making it perfect for real-world implementation. With the help of contemporary architectural designs, Xception (94.96%) and InceptionV3 (94.68%) came in close succession. Although they needed more processing power, DenseNet121 (94.24%) and ResNet50 (93.96%) also did well. VGG16, on the other hand, demonstrated low accuracy (35.97%), most likely as a result of overfitting brought on by its outdated design and undeveloped feature extraction. All things considered, NASNetMobile was the most resource-efficient and accurate model for detecting lung diseases.

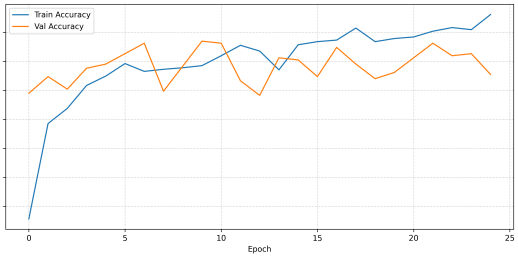


Fig. 4. Accuracy curves of the proposed NASNetMobile model.

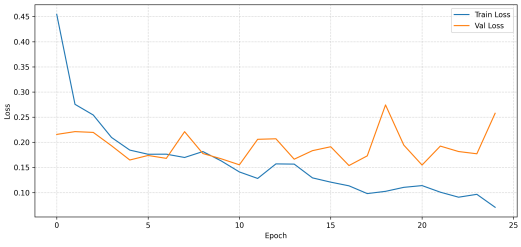


Fig. 5. Loss curves of the proposed NASNetMobile model.

Lung opacity, normal, and viral pneumonia are the three classes that the proposed NASNetMobile-based lung illness

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classification model can discriminate between, as shown by the ROC curves in Figure 6. The model achieves impressive Area Under the Curve (AUC) values of 0.98 for both the Lung Opacity and Normal classes, and a perfect AUC of 1.00 for the Viral Pneumonia class. These near-ideal ROC curves indicate a high true positive rate with very few false positives across all classes, demonstrating the model's excellent discriminative power and reliability in correctly identifying lung diseases from chest X-ray images.

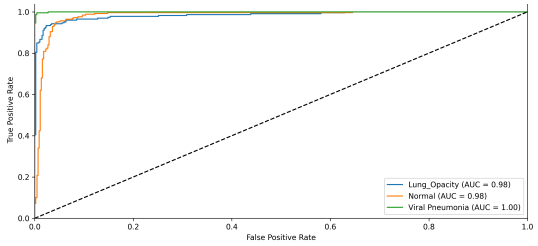


Fig. 6. ROC curve for the proposed NASNetMobile model.

The classification performance of the proposed NASNetMobile model for lung disorder identification from chest X-ray images is illustrated by the confusion matrix shown in Fig. 7. The model accurately classified 238 normal cases, 219 viral pneumonia cases, and 206 lung opacity cases, highlighting its strong discriminative ability. Misclassifications were minimal, with 9 normal cases misclassified as lung opacity, 3 normal cases as viral pneumonia, and 1 viral pneumonia case misclassified as normal. The significant diagonal dominance in the confusion matrix confirms the model's high accuracy and reliability, establishing NASNetMobile as an efficient lightweight architecture for automated lung disease diagnosis.

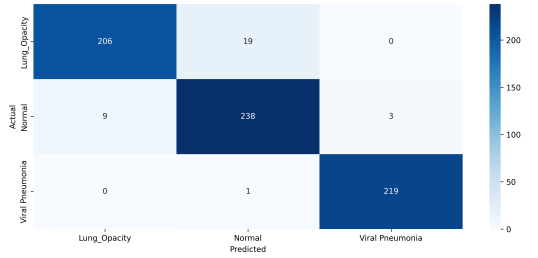


Fig. 7. Confusion matrix of the proposed NASNetMobile model.

Table V summarizes the classification performance of the proposed NASNetMobile model for lung disease detection. The model achieves very high precision (95.8%–99.6%) and recall (95.2%–100%) across the three classes—Normal, Viral Pneumonia, and Lung Opacity—resulting in strong F1-scores above 97% for all categories. The perfect recall for Lung Opacity indicates no missed cases, while high precision values show minimal false positives. With balanced support for each class, these metrics confirm the model's robust and reliable capability in accurately identifying lung disorders from chest X-ray images, making NASNetMobile a highly effective lightweight architecture for automated diagnosis.

TABLE V
CLASSIFICATION METRICS OF PROPOSED MODEL

Class	Precision (%)	Recall (%)	F1-score (%)	Support
Normal	99.6	95.2	97.4	250
Viral Pneumonia	98.6	99.5	99.0	220
Lung Opacity	95.8	100.0	97.9	206

Significant performance variations across several deep learning architectures for the categorization of lung diseases are highlighted by the model accuracy comparison that is provided. With an accuracy of 95.40%, the suggested model NASNetMobile marginally outperformed Xception (94.96%) and InceptionV3 (94.68%). Additionally, with accuracies of 94.24% and 93.96%, respectively, DenseNet121 and ResNet50 both showed competition. On the other hand, VGG16's low accuracy (35.97%) suggests architectural incompatibility (e.g., lack of skip connections) or suboptimal hyperparameters., drastically behind the other models. Potential problems with VGG16's implementation, dataset compatibility, or inadequate training setup are suggested by this disparity. Overall, the findings suggest that while earlier models like VGG16 could need considerable modification to reach equivalent performance, more recent, lightweight architectures like NASNetMobile and Xception are better suited for lung disease detection applications.

TABLE VI
MODEL ACCURACY COMPARISON FOR LUNG DISEASE CLASSIFICATION

Model	Accuracy/Performance (%)
VGG16	35.97
ResNet50	93.96
DenseNet121	94.24
InceptionV3	94.68
Xception	94.96
Proposed NASNetMobile	**95.40**

The classification accuracy of many deep learning models employed for lung disease detection is compared in Table VII. The suggested NASNetMobile model performs better than current methods with a 95.40% accuracy rate, proving its efficacy and resilience in the categorization of multiclass lung diseases. Harshavardhan et al. [17], on the other hand, stated that DenseNet-121 had an accuracy of 88.00%, which is significantly lower. According to Liu et al. [18] and Irtaza et al. [19], MobileNet-Lung and MobileNetV1 both achieved competitive accuracies of 93.30% and 93.40%, respectively. These outcomes demonstrate the NASNetMobile model's higher performance, which was established in this study.

TABLE VII
ACCURACY COMPARISON OF LUNG DISEASE CLASSIFICATION MODELS

Model	Accuracy (%)	Source
NASNetMobile	**95.40**	This work
DenseNet-121	88.00	Harshavardhan *et al.* [17]
MobileNet-Lung	93.30	Liu *et al.* [18]
MobileNetV1	93.40	Irtaza *et al.* [19]

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This figure illustrates an Explainable AI (XAI) visualization using Gradient-weighted Class Activation Mapping (Grad-CAM) to interpret a CNN's decision process (Fig. 8). It consists of three parts: the original canine image, the Grad-CAM heatmap (cool-to-warm colors showing low-to-high saliency), and the superimposed result. The highlighted facial regions indicate that the model's classification of "R" (Rottweiler) was mainly influenced by discriminative features in the head, thus enhancing transparency and validating the model's reasoning.

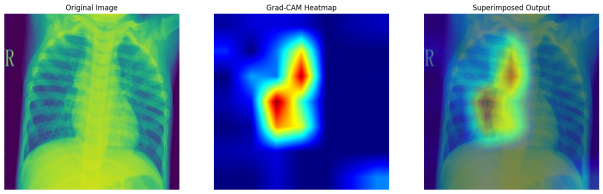


Fig. 8. Grad-CAM Visualization for the proposed model.

VII. Conclusion

In order to categorize chest X-ray pictures into three groups—lung opacity, normal, and viral pneumonia—this study included a detailed comparison of many pre-trained deep learning models. Xception (94.96%), InceptionV3 (94.68%), DenseNet121 (94.24%), and NASNetMobile (95.40%) were the models that performed the best in terms of classification accuracy among the models that tested. According to these findings, even deeper models can be outperformed while preserving computing efficiency by using powerful and lightweight architectures like NASNetMobile. In contrast, VGG16's accuracy of only 35.97% demonstrated its considerable underperformance, underscoring its limits in managing intricate multi-class classification tasks on this dataset in the absence of further tuning or architecture improvements.

Although the outcomes are promising, several limitations exist. First, despite its good structure, the dataset might not accurately depict clinical situations with diverse patient populations and changing image quality. Although class weights are used to remedy the imbalance, model generalization may still be impacted. Additionally, clinical metadata (such as patient history and symptoms) that could improve diagnostic accuracy is absent from the current investigation, which exclusively uses image-based inputs. Furthermore, the subpar performance of VGG16 highlights the possibility that extra fine-tuning, layer freezing techniques, or data-specific adaptation may be necessary for some models to produce competitive outcomes.

Future work should integrate multimodal data (e.g., lab tests, patient history) to enhance diagnostic accuracy and clinical utility. Grad-CAM and other model interpretability and explainability techniques should be used to promote therapeutic trust and openness. The evaluation of latency and inference time on edge devices could be done by simulating real-time deployment scenarios. Moreover, incorporating hybrid models or ensemble learning techniques may further enhance accuracy. Finally, increasing the dataset's diversity by including more samples from underrepresented groups or uncommon diseases will improve its resilience and suitability for use in actual healthcare settings.

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